

Forces and motion

It can be difficult to make an object move, but once it is moving, it will continue to move until something stops it. Force is needed to start something moving, make it move faster, and make it stop.

The soccer ball would stay still if the player didn't kick it.



Newton's laws of motion

In 1687, Isaac Newton presented three important rules that explain how forces make things move. They have become the foundation of physics and work for just about everything, from soccer balls to frogs.



Newton's second law

The bigger the force and the lighter the object, the greater the acceleration. A professional cyclist with a lightweight bike will accelerate faster than a normal person cycling to work.

Forces make things accelerate. In this case, the force is created by the cyclist's powerful legs.

Newton's first law

An object stays still, or keeps moving in a straight line at a constant speed, if it isn't being pushed or pulled by a force.



Newton's third law

Every action has an equal and opposite reaction. The leaf moves away as the frog leaps in the opposite direction.